

Audion Voice AI - User Guide

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This guide describes the current workflow for Audion Voice AI Live and Audion Voice AI Studio.

1. Choose an Edition

Use **Audion Voice AI Live** when you need a compact distribution with API Live and local models. This is the primary laptop and daily-use build.

Use **Audion Voice AI Studio** when you have a CUDA workstation and need fast local large-model processing, faster-whisper, PyTorch, and GPU diarization.

2. Installation

Installing the recognition models

The build ships without model weights: the GigaAM cache and the whisper.cpp pack come to about 3.3 GB together. Weights carry their own licences, apart from the licence of this program, so they are downloaded by you — under your own account where that is required.

Until the models are installed, transcription will not run: the window opens, but there is nothing behind it to recognise speech.

On the first start the app offers to download what is missing: the "Download models and engines" window lists the modules with their download size, for Live that is GigaAM and whisper.cpp with the Turbo model, about 3.3 GB. "Download and install" installs them one after another; progress is shown on the Maintenance page, and once everything is in place every mode there is green. "Later" postpones the question until the next start, "Don't ask again" hides the window for good. Modules can still be installed by hand on the same page.

The same can be done by hand: run `builder_main.cmd` and pick, in order:

#	Menu entry	What it installs
08	<code>GIGAAM ONNX</code>	the main Russian speech recognition model
09	<code>WHISPER.CPP CPU FALLBACK</code>	the CPU fallback engine

Both steps are required. The download comes from HuggingFace and takes as long as several gigabytes usually take.

Open `builder_main.cmd` or the `Maintenance` tab in the GUI.

Recommended order:

1. Start the app. On the first start the "Download models and engines" window installs everything recommended with one button; the remaining steps are for manual repair or checks.
2. `builder_main.cmd` is only needed when the app is not built yet: it checks the folder structure and installs the Python runtime.
3. FFmpeg ships with the distribution. `Reinstall` on the `Maintenance` page is only needed after an NVIDIA driver change, because the build is picked to match the driver.
4. `Dependency wheel cache` (`wheelhouse`) ships with the distribution and shows as `Installed`. `Reinstall` is only needed if the folder was deleted: it downloads the GigaAM/ONNX Runtime wheels again.
5. Live dependencies (microphone, streaming dictation) are installed by the GUI itself at start-up from `wheelhouse\live`, without Internet access. The manual row remains available for repair.
6. Click `Check` in the `Microphone check` card: it tests the Windows default recording device, then a separate communications default, including native 44.1/48 kHz modes. Audio is not saved.
7. `GigaAM ONNX pack`: onnx-asr, the ONNX Runtime provider, the GigaAM v3 models and Silero VAD for splitting long recordings.
8. `whisper.cpp pack`: Live installs the CPU build, Studio the CUDA/cuBLAS build; the Turbo model comes with it.
9. Studio: `whisper.cpp Large V2 model` is the primary file model in CUDA mode. `GPU diarization` (torch + pyannote Community-1) is optional, NVIDIA only and needs a HuggingFace key; the first-start window never installs it.
10. `Speaker separation (no key)`: sherpa-onnx and two models, about 65 MB, optional; the first-start window does not install it because the separation is coarse.
11. Run base verify/smoke checks.

The Maintenance tab shows the detected GPU, recommended profile, progress, speed, and ETA.

Installer output is routed automatically to the left `Activity log`; there is no empty terminal area on the right. Rows are marked `Recommended`, `Optional`, or `Not needed`; non-recommended actions remain legible and available.

`builder_main.cmd` and the GUI use the same install scripts. The builder is for the first portable build, recovery, and manual maintenance; the GUI repairs the small Live dependency set automatically, while other modules remain available in `Maintenance`.

3. API Keys

API modes read keys from `config`.

```
config/  
api_key_*.txt
```

Reset App does not delete these files. Cleanup must also protect working configs.

The **HuggingFace key** is separate and optional (`config\api_key_huggingface.txt`, the "HuggingFace key" row on the `Maintenance` page). Without it the models download anonymously, only slower; with a key HuggingFace lifts the rate limit. Getting one takes a minute: huggingface.co → sign in or sign up → avatar → Settings → Access Tokens → New token, type Read → copy and paste it into the key row. In Studio the same key unlocks speaker separation: with the same account open the `pyannote/speaker-diarization-community-1` page and accept the model terms.

4. Main Window

Main tabs:

- `Live` - dictation, overlay, API/Local sources, models, and OpenAI cleanup.
- `Files` - file queue, models, transcription language, post-processing/cleanup, subtitles, and export.
- `Settings` - theme, app language, tray, Notion/Obsidian integrations, global options, and settings reset.
- `Maintenance` - recommended profile, runtimes, payloads, models, and service actions.

The left side contains the log and queue. The right side contains workflow settings. Inactive backend cards are hidden or dimmed so API, Local Models, and CUDA settings are not confused.

Right-click the compact top overlay for an 85%-opaque quick menu. Its first actions are `Past dictations` and `Paste last dictation`, followed by Live dictation mode, file recording mode, `Input` / `Output`, the main window, Settings, and Exit. The quick overlay window shows the latest 20 entries. Tray `All dictations` opens the scrollable cache of up to 200 completed Live dictations; the oldest entries are removed automatically at the limit. Click a two-line card to paste it back into the previously active window; use its compact round buttons to copy or delete it. In Live mode the ready overlay shows the microphone; in file mode it shows a red Record symbol and `Start recording`. During a file recording, Pause omits that interval from the WAV while keeping the microphone open,

Stop saves, and Cancel discards the temporary recording. The overlay and tray have no tooltips because every action is labeled or uses standard transport symbols.

The default overlay scale for a fresh installation is 70%. Later changes are saved. The mouse wheel scrolls the Settings page without changing the scale value under the pointer.

The **Right Alt + F12** hotkey is activated automatically with the application. Local STT has a separate **Preload local model** option: it speeds up the first dictation but reserves RAM/VRAM in advance. It does not control the overlay, tray, or hotkey readiness.

5. App Language and Transcription Language

These are separate settings.

- **App language** changes the interface.
- **Transcription language** tells the engine what language is spoken in the recording.

Use **Auto** when the language is unknown. If the recording is definitely Russian or English, choose the exact language.

Live dictation has a separate **Primary language**. Its default is **Window layout**: when dictation starts, the app reads the current Windows keyboard layout of the actual target window where text will later be pasted (**ru** or **en**). Russian, English and Auto explicitly override it. The shared **Term dictionary** is visible on both the **Live** and file-operation tabs in every API/Local mode. It contains only exact spellings of names, abbreviations and formats, separated by commas. The dictionary is sent to Live and file STT as provider keyterms/prompt where the selected engine supports prompting, and protects spellings during optional transcript formatting. GigaAM ONNX accepts neither prompts nor hotwords, so the dictionary does not change its raw recognition. The separate **Recording description** field is free-form context about the topic, participants, purpose and important facts; the dictionary does not need to be repeated there.

6. File Transcription

1. Open **Files**.
2. Add files or a folder with the separate round picker buttons beside **Add...**. Use the checkboxes in the first queue column to target particular files. If no rows are checked, Start processes the complete queue; bulk remove/cleanup actions still fall back to ordinary row selection.
3. Choose the file engine: OpenAI, Local Models, or CUDA in Studio.
4. Choose the transcription language.

5. For OpenAI, choose a transcription profile; for local processing, choose the model and backend.
6. Enable required export formats.
7. Start processing.

Results are saved next to the source file by default. This is better for large archives because the transcript stays with the original recording.

A folder is added with all its subfolders: supported files join the queue as a flat list and each transcript is saved next to its file in that folder. The full format list (17 audio and 18 video) is shown in the tooltips of `Files...`, `Folder...` and the `Formats` label. External files selected through the picker or drag-and-drop are not copied into the project. The queue retains their original paths and processes them in place. Only WAV files recorded by Audion itself are created automatically in `input`. The `Input` and `Output` buttons share the remaining width and open those project folders.

7. Supported Formats

The app relies on FFmpeg, so common audio and video formats are preferred:

- audio: WAV, MP3, M4A, AAC, FLAC, OGG, OPUS;
- video: MP4, MOV, MKV, WEBM, AVI.

If a file uses an exotic container, convert it to WAV, MP3, or MP4 first.

8. API Workflow

OpenAI is the fastest path for high-quality cloud transcription and text cleanup. In Live, the OpenAI model is not selected manually: Realtime uses `gpt-realtime-whisper`, and batch fallback uses `gpt-4o-mini-transcribe`. xAI and ElevenLabs are currently used as fixed realtime Live providers.

Important settings:

- valid API key for the selected provider;
- OpenAI file transcription profile;
- post-processing model;
- cleanup prompt when text cleanup is enabled.

OpenAI file profiles:

- `Fast / economical` -> `gpt-4o-mini-transcribe`;
- `Max accuracy` -> `gpt-4o-transcribe`;

- `With diarization` -> `gpt-4o-transcribe-diarize`.

Each profile has a tooltip explaining the tradeoff. The model refresh button remains useful for post-processing and cleanup models.

9. Local Models Workflow

Local Models keep audio out of external APIs. They are useful for private work, long recordings, and machines with an installed GPU/CPU backend.

Local models:

- GigaAM - preferred local choice for the Russian UI layout;
- whisper.cpp - CPU fallback in Live and CUDA/cuBLAS GPU pack in Studio;
- backend: auto, CUDA, DirectML, or CPU fallback;
- GigaAM Live keeps the model warm until the app exits;
- GigaAM cuts long recordings into pieces of up to 25 seconds at the quietest points (by Silero VAD, installed with the GigaAM ONNX pack; by signal energy without it). Nothing is discarded, quiet echoing speech still reaches the model, and every piece gets start and end times;
- unload/buffer threshold applies to whisper.cpp live scenarios.

Install the required runtimes/payloads and models from the `Maintenance` page before using this mode (or accept the first-start window). The GigaAM/ONNX Runtime wheels ship in the distribution's `wheelhouse`: `GigaAM ONNX pack` installs `onnx-asr` and the ONNX Runtime provider from them, preloads `gigaam-v3-e2e-ctc` / `gigaam-v3-e2e-rnnt` into `models\huggingface` and fetches Silero VAD. On Windows, auto uses DirectML as the lightweight universal backend; Studio uses CUDA on NVIDIA.

Speaker separation in local modes is switched on by the `Speaker separation` card on the `Files` tab (the same state as the `Transcription + diarization` fork in API mode). The path without a HuggingFace key is installed by the `Speaker separation (no key)` module: sherpa-onnx and two ONNX models, about 65 MB, runs on the CPU and takes roughly a fifth to a half of the recording's length. This engine needs the number of speakers in the `Speakers` field (2 to 20). It has no automatic mode on purpose: sherpa-onnx has no calibrated threshold, on one and the same recording it finds anywhere from one to seven people depending on the setting, and we chose not to tune the app to a single recording. Without a number separation does not run, the transcript is saved without labels, and the journal keeps a warning. Separation without a key is coarse: it splits unlike voices and may merge similar ones. On a test recording of three people in one room it agreed with Community-1 on three

quarters of the segments and took two similar voices for one; Community-1 found all three on the same recording with no hint.

9.1. Local Backend Installation

Audion chooses the backend by actually loading provider DLLs/runtimes, not just by GPU name.

- **DirectML**: the lightweight Windows fallback for Live and a reserve path for Studio. **GigaAM ONNX pack** installs it as **onnxruntime-directml**; no external SDK is required. Keep the NVIDIA/AMD/Intel driver current.
- **CUDA**: the NVIDIA path for Studio. Install the current NVIDIA driver and Studio runtime/payloads from **Setup** / **builder_main.cmd**, then run **GigaAM ONNX pack**. ONNX Runtime uses **onnxruntime-gpu**; CUDA/cuDNN/MSVC DLLs must be visible to the process, and Audion also calls **onnxruntime.preload_dlls()**.
- **TensorRT**: not used in the current project profile. If ONNX Runtime exposes a TensorRT provider, Audion does not select it as a recommended backend.

10. CUDA Workflow in Studio

CUDA is available only in Studio. It targets NVIDIA GPUs and covers GigaAM ONNX CUDA, whisper.cpp CUDA/cuBLAS, and faster-whisper/pyannote.

The **CUDA** card includes a Faster-Whisper profile switch:

- **Quality** - the default mode. It uses regular Faster-Whisper/CTranslate2 without batched inference. GPU load is calmer, and it is safer for rough speech, quiet intros, production chatter, and later diarization because the timeline is usually more detailed.
- **Speed** - enables BatchedInferencePipeline with **batch_size=16**. This mode keeps CUDA/GPU much busier and speeds up long files or file queues, but it depends more on VAD, can merge larger segments, and can occasionally miss quiet boundary phrases. Use it when you need a fast run and can dedicate the GPU to the job.

Typical flow:

1. Install **whisper.cpp pack** (CUDA/cuBLAS) and **whisper.cpp Large V2 model**; the first-start window does this by itself.
2. When speaker labels are needed: enter the HuggingFace key on the **Maintenance** page, accept the terms of **pyannote/speaker-diarization-community-1**, and install **GPU diarization**. Without a key the install does not start and says why.

3. Run verify.
4. Turbo remains the fast profile for comparisons against large-v2.
5. Choose the Studio GPU engine: GigaAM CUDA, whisper.cpp cuBLAS, or faster-whisper CUDA.
6. Run a short smoke test on a small file.
7. Process long recordings after the smoke passes.

RTX 5070 smoke confirmed the target Studio GigaAM CUDA and Studio whisper.cpp CUDA/cuBLAS profiles. On a machine without NVIDIA GPU, CUDA smoke can only check imports.

11. Live Dictation

Live dictation can be started from the **Live** tab, the record button above the log, or the tray.

Modes:

- **API models** - OpenAI batch, OpenAI Realtime, xAI Realtime, or ElevenLabs Realtime. The OpenAI model is selected by the app and is not exposed as a catalog.
- **Local Models** - GigaAM or whisper.cpp with the selected backend.

Choose the source first (**API models** or **Local Models**), then choose the concrete provider or local model. The inactive card remains dimmed so the screen does not feel empty while still avoiding accidental configuration changes.

When **On startup** is enabled, dictation starts with the app.

12. Overlay

After startup, the overlay remains at the top and reserves the full controller's 420-680 px native width. In idle, a window mask leaves only the centered 120×12 px oval visible and interactive. Hover clears the mask and immediately reveals the full-width Record capsule with a stationary microphone icon at its exact center. Clicking fades the icon over roughly 220 ms and swaps in the recording controls inside the same bounds without resizing or moving the window. The minimum/default height is 52 px. Stop, Cancel, or the safety timeout returns it to the thin oval. It disappears only when the application exits or overlay display is disabled.

The left **Activity log** stays empty until dictation, an error, or a maintenance operation occurs. The armed state and **Right Alt + F12** reminder are shown in the microphone tooltip and status bar instead of being written as startup log entries.

The safe global start hotkey is **Right Alt + F12**. It contains no Windows key, so Windows Search cannot steal the paste target.

The right side of the overlay contains an elapsed-time counter and two controls:

- **Stop** ends recording, waits for the final STT result, pastes the complete text, and then closes the overlay;
- the left **×** cancels the session and discards its collected text. Streaming text remains in the overlay and is not pasted into the target application before **Stop**;
- **Stop** pastes the complete result as one block. A configurable safety interval (15 minutes by default) finalizes a forgotten continuous recording automatically.

The overlay accepts clicks on these controls without taking keyboard focus away from the dictation target.

The actual microphone level is shown to the left of the text, while cloud transport state appears on the right (**connecting**, **ready**, **sending audio**, **receiving text**). Long text is elided on the left according to its rendered width, keeping the latest dictated words visible.

The far-left **:** kebab is the drag handle and is already visible beside the ready microphone before recording starts. Move the overlay without starting voice capture; the selected position is reused for subsequent appearances until the application exits.

The **Live** tab allows you to:

- enable or disable the overlay;
- adjust overlay height;
- control overlay behavior during live dictation.

13. Live Text Cleanup

Enable **Live Cleanup** for long dictation sessions.

Default cleanup is conservative and punctuation-first: it restores punctuation, capitalization, sentence boundaries and paragraphs without paraphrasing, reordering ideas or replacing words with synonyms. Only obvious fillers, false starts and accidental repetitions may be removed.

Settings:

- cleanup model;
- cleanup prompt;

- sentence count threshold.

The app can periodically turn raw dictated text into a cleaner document.

14. Export

Export is available from the GUI and tray.

Formats:

- Markdown;
- TXT;
- JSON;
- SRT;
- WebVTT.

Actions:

- save the log to Markdown through a file picker;
- export a transcript;
- send material to Notion;
- send material to Obsidian.

15. Tray

The tray is for quick actions while the main window is hidden.

Minimizing the window hides the app in the tray. Closing the window exits the app completely, stops background work, and removes the tray icon.

Available actions:

- show or hide the app;
- start or stop live dictation;
- export the log to Markdown;
- send results to Notion or Obsidian.

Tray behavior can be disabled in [Settings](#).

16. Settings Persistence

The app saves the following between restarts:

- theme;
- app language;
- checkboxes;
- filled fields;
- selected models;
- model lists;
- live settings;
- export settings;
- tray, integrations, and global app behavior settings.

Combo boxes do not change on mouse wheel, so accidental scrolling cannot alter workflow settings.

17. Reset App

Reset App restores default program settings. Use it when the interface or workflow was accidentally configured into a bad state.

Reset App must not delete:

- API keys;
- installed runtimes;
- payloads;
- downloaded models;
- user work files.

18. Cleanup

cleanup_project.cmd removes what can be restored on another system: temporary build artifacts, runtime, **Tools**, models, **install\download**, **install\wheels**, and working payload folders.

It also clears **input**, **output**, **logs**, **report**, **workspace**, and **release**. This is expected:

input is a temporary working area for files, not a user archive.

Working configs, API keys, install scripts, `system_core`, `Docs`, `tests`, and important root project files are protected. After cleanup, the empty structure is recreated through `install\init_folders.cmd`.

The GUI module catalog and `builder_main.cmd` are synchronized. The target profiles were smoke-tested on RTX 5070: Live GigaAM DirectML, Live whisper.cpp CPU fallback, Studio GigaAM CUDA, and Studio whisper.cpp CUDA/cuBLAS.

19. Recommendations

- Use API models for short and simple jobs.
- Use Local Models for private or long local work.
- Use Studio and CUDA for large archives on NVIDIA GPUs.
- Save output next to the source file.
- Do not confuse app language with transcription language.
- Run a short smoke test before processing multi-hour recordings.

20. Troubleshooting

- Check the log on the left.
- Make sure FFmpeg is installed.
- Check API keys for OpenAI, xAI, or ElevenLabs.
- For Local Models, check the model and runtime.
- For CUDA, check the NVIDIA driver and that the CUDA/cuBLAS `whisper.cpp pack` with the Large V2 model is installed; PyTorch is only needed for diarization.
- Long GigaAM recordings are cut into pieces of up to 25 seconds automatically. If a piece still fails with a DirectML error, update the GPU driver or switch the backend to CPU.
- If speaker separation fails (pyannote missing, no HuggingFace key, model terms not accepted), the transcript is still saved and the journal shows `WARN: speaker separation skipped` with the reason and a hint.
- If models were installed by hand and the `Mode readiness` matrix is not green, open `Maintenance`: the row marked `Not installed` shows what is missing.
- Use Reset App if the issue looks like broken UI configuration.

Live Session Checklist

Select the correct microphone, input language, recognition engine, and text destination before starting. Record a short sample and listen for clipping, silence, wrong device routing, or excessive background noise. Confirm that live text appears in the intended window and that stop releases the device.

For overlay use, verify screen position, focus behavior, opacity, and click-through settings before a production session. The overlay must not hide an error or continue displaying stale text after recognition stops.

Local And API Boundaries

API engines require network access, a valid provider key, and the selected model. Local engines require the installed backend, model payload, and supported CPU/DirectML/CUDA profile. Do not diagnose a local model failure by changing an unrelated API key.

When privacy requires offline processing, confirm that the selected route is local and that no provider fallback is enabled. Logs and exported transcripts may contain sensitive speech and should follow the same retention rules as source audio.

Long Sessions

Check free disk space, output folder, model memory usage, and power settings. Keep the system awake and prevent overlapping capture applications from taking exclusive control of the device. Review segment boundaries and timestamps periodically rather than waiting until the end of a long recording.

Failure Recovery

Preserve the log, selected engine/model, device, language, and a short reproducible sample. Restart only the failed session or backend when possible. Use Reset App only for corrupted UI/settings state; it is not a substitute for reinstalling a missing runtime or model.

Interface As Documentation

Where the project exposes declarative GUI settings or configuration maps, treat them as structured documentation sources for controls, defaults, tooltips, and backend choices. The guide adds the

human sequence: select the correct microphone and language, understand whether audio leaves the machine, monitor a long session, and preserve recoverable output.

For release acceptance, compare the rendered interface, persisted settings, actual engine request, export result, and this guide. A new control must explain its effect on latency, quality, privacy, memory, or output rather than existing only as a label.